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NUMERICAL METHODS FOR THE JIN-XIN MODEL

IMEX SCHEMES AND RELAXNN FRAMEWORKS

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Kinetic Equations: Classical Foundations and Modern Computational Perspectives Examination
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Introduction & Mathematical Background

Implicit-Explicit Methods & Relax Neural Networks

Implicit Explicit Neural Network (Proposal)

REFERENCES

- [1] S. Jin and Z. Xin, “**The relaxation schemes for systems of conservation laws in arbitrary space dimensions,**” *Communications on Pure and Applied Mathematics*, 1995.
- [2] U. M. Ascher, S. J. Ruuth, and R. J. Spiteri, “**Implicit-explicit runge-kutta methods for time-dependent partial differential equations,**” *Applied Numerical Mathematics*, 1997.
- [3] S. Boscarino, L. Pareschi, and G. Russo, *Implicit-Explicit Methods for Evolutionary Partial Differential Equations*. Society for Industrial and Applied Mathematics, 2024.
- [4] N. Zhou and Z. Ma, ***Capturing shock waves by relaxation neural networks***, 2024.

INTRODUCTION & MATHEMATICAL BACKGROUND

THE TARGET: BURGERS' EQUATION AND SHOCK FORMATION

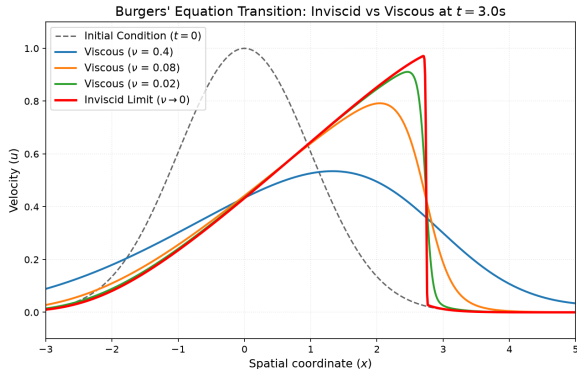
■ Inviscid Burgers' Equation:

$\partial_t u + \partial_x (f(u)) = 0$, where f is a differentiable convex function.

- If $f'(u_0)$ is not a weakly increasing function, a sharp shockwave develops.

■ Viscous Burgers' Model:

$\partial_t u + u \partial_x u = \nu \partial_{xx} u$ introduces a parabolic regularization, yielding smooth profiles.



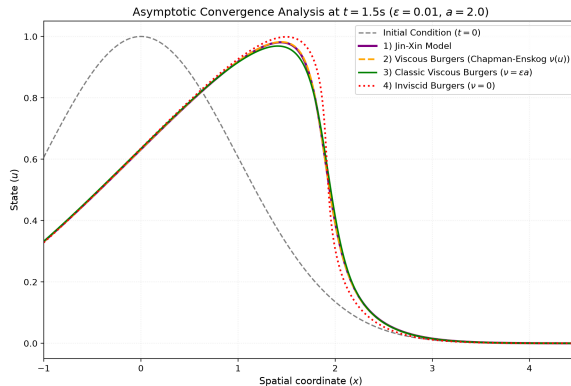
Linear Hyperbolic System with Stiff Relaxation

$$\begin{cases} \partial_t u + \partial_x v = 0 \\ \partial_t v + a \partial_x u = -\frac{1}{\varepsilon}(v - f(u)) \end{cases}$$

Key Mathematical Properties

- **Characteristic Speeds:** Information propagates at fixed linear velocities $\pm\sqrt{a}$ (Riemann Invariants).
- **Sub-characteristic Condition:** Linear stability requires $a > (f'(u))^2$ to avoid unphysical growing modes.

ASYMPTOTIC EQUIVALENCE: CHAPMAN-ENSKOG EXPANSION



Chapman-Enskog Expansion ($\varepsilon \rightarrow 0$):

$$\partial_t u + \partial_x f(u) = \partial_x \left[\varepsilon \left(a - (f'(u))^2 \right) \partial_x u \right] + \mathcal{O}(\varepsilon^2)$$

Linearized Transport

- Replaces complex non-linear fluxes with straight, linear characteristic fields.

Natural Dissipation

- Physical viscosity is implicitly embedded via the algebraic relaxation scale ε .

Shock Smoothing

- Overcomes classical *spectral bias* pathologies using trainable, smooth profiles.

First-Order Autodiff

- Relies strictly on highly efficient **first-order backpropagation** paths.

IMPLICIT-EXPLICIT METHODS & RELAX NEURAL NETWORKS

$$\begin{cases} \partial_t u + \nabla_x \cdot \mathbf{v} = 0 \\ \partial_t \mathbf{v} + a \nabla_x u = -\frac{1}{\varepsilon} (\mathbf{v} - \mathbf{f}(u)) \end{cases}$$

- **The Classical Bottleneck:** Conventional Finite Volume and IMEX schemes scale poorly in high-dimensional or multi-parametric settings (*Curse of Dimensionality*).
- **Mesh Dependency:** Standard discretizations require continuous grid regeneration and are bounded by tight, resolution-dependent CFL constraints.
- **The SciML Paradigm Shift:** Neural Operators act as **mesh-free, continuous maps**. Once optimized, inference achieves $\mathcal{O}(1)$ speed, invariant to spatial resolution, parameter spaces (ε, a) , or query times.

Operator Splitting for Stiff Systems

The Jin-Xin formulation exhibits multi-scale dynamics: a non-stiff linear convective transport coupled to a highly stiff algebraic relaxation source term ($\varepsilon \ll 1$).

- **Convection (Explicit Step):** Update the macroscopic state u^{n+1} by resolving the linear advection terms explicitly:

$$u^{n+1} = \text{ExplicitScheme}_{\Delta t} (\nabla_x \cdot \mathbf{v}^n \mid u^n)$$

- **Relaxation (Implicit Step):** Update the relaxation variable $\mathbf{v}^{n+1}$ by treating the stiff algebraic source implicitly:

$$\mathbf{v}^{n+1} = \text{ImplicitScheme}_{\Delta t} \left(a \nabla_x u^n - \frac{1}{\varepsilon} (\mathbf{v}^{n+1} - \mathbf{f}(u^{n+1})) \mid \mathbf{v}^n \right)$$

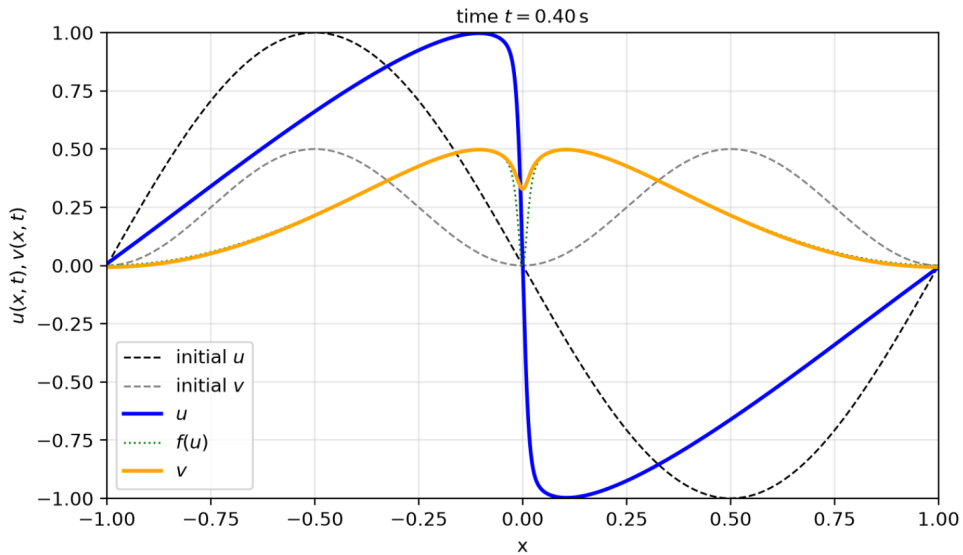
- **Spatial Gradients:** The linear operators $\nabla_x \cdot \mathbf{v}^n$ and $a \nabla_x u^n$ are discretized via a conservative, non-oscillatory Rusanov numerical flux.

- **Explicit Forward Euler Step:**

$$u^{n+1} = u^n - \Delta t \nabla_x \cdot \mathbf{v}^n$$

- **Implicit Backward Euler Step:**

$$\mathbf{v}^{n+1} = \frac{\varepsilon \mathbf{v}^n - \varepsilon \Delta t a \nabla_x u^n + \Delta t \mathbf{f}(u^{n+1})}{\varepsilon + \Delta t}$$



- **Chapman-Enskog Inspired Decoupling:** Introducing the micro-fluctuation variable $\mathbf{w}$ via $\mathbf{v} = \mathbf{f}(u) + \varepsilon \mathbf{w}$ transforms the governing equations into:

$$\begin{cases} \partial_t u + \nabla_x \cdot \mathbf{f}(u) = -\varepsilon \nabla_x \cdot \mathbf{w} \\ -\partial_t \mathbf{f}(u) - a \nabla_x u = \mathbf{w} + \varepsilon \partial_t \mathbf{w} \end{cases}$$

- **Asymptotic Preserving Loss ($\varepsilon \rightarrow 0$):** Forces convergence to the local equilibrium manifold $\tilde{\mathbf{v}} = \mathbf{f}(\tilde{u})$ via:

$$\mathcal{L}_0(\tilde{u}, \tilde{\mathbf{w}}) = \text{MSE}(\partial_t \tilde{u} + \nabla_x \cdot \mathbf{f}(\tilde{u}), 0) + \text{MSE}(-\partial_t \mathbf{f}(\tilde{u}) - a \nabla_x \tilde{u}, \tilde{\mathbf{w}})$$

Remark: Standard un-decoupled networks lack this structural ε -shield, causing high-frequency oscillations and requiring prohibitive training budgets ($\sim 300\text{k}$ epochs).

- **Network Topology:** A shared deep **Encoder** with SiLU activations extracts the spacetime embedding, branching into decoupled **Heads** for the macro ($\tilde{u}$) and micro ($\tilde{w}$) fields.
- **Geometric Shock Shaping:** Deploying Tanh activations in the final layer mimics the exact analytical profile of a viscous traveling shock wave:

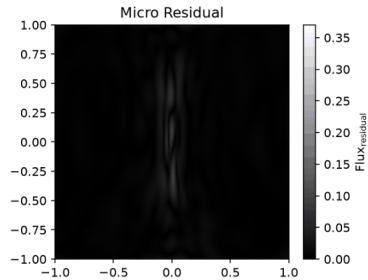
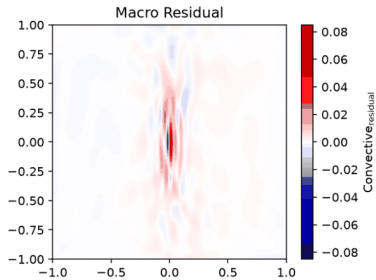
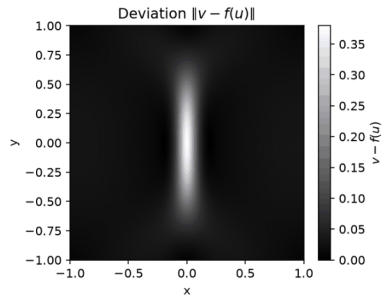
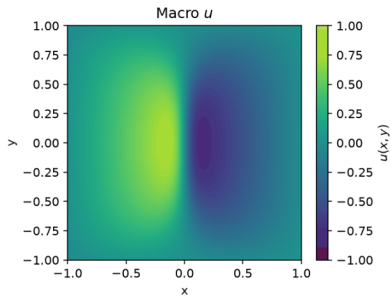
$$u(\mathbf{x}, t) = \frac{u_L + u_R}{2} - \frac{u_L - u_R}{2} \tanh \left(\frac{(u_L - u_R)(\mathbf{x} - \mathbf{s}t)}{4\nu} \right)$$

- **Gradient Saturation Prevention:** While Tanh constrains the final decoding space to bound physical discontinuities, the deep SiLU encoder prevents early gradient freeze.

- **Loss Formulations:** MSE loss tracks boundary and initial data interpolation; MAE ($\mathcal{L}_1$) loss minimizes physics residuals to enforce sharp shock capturing. L_1 weight regularization targets the heads, and L_2 regularizes the encoder.
- **Batch Dimensioning:** To counter dimensional degradation, sampling scales according to the covering framework:

$$N = C_d \frac{V}{\epsilon} \left(\ln \frac{V}{\epsilon} + c \right) = \mathcal{O}(V \ln V)$$

where V represents domain volume, ϵ is elementary volume, c denotes covering confidence, and C_d embeds geometric dependencies.



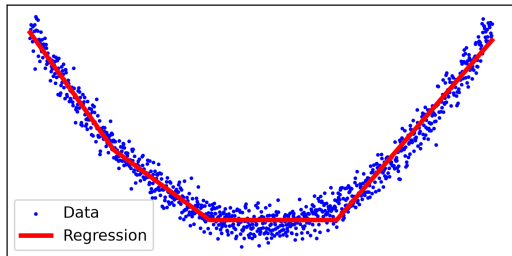
IMPLICIT EXPLICIT NEURAL NETWORK (PROPOSAL)

MOTIVATION AND THE CORE IDEA

The Challenge: Hyperbolic conservation laws resolved on **coarse grids** incur severe sub-grid information loss. Classical stencils (e.g., Rusanov) trade off excessive numerical diffusion against spurious unphysical oscillations.

Objective: Optimize *only* the discrete spatial differentiation operator via data, embedding it within a conservative, stable, and well-understood IMEX time-stepping framework.

- Circumvents uninterpretable, black-box temporal trajectory tracking.
- Operates as a data-driven spatial regression function, structurally bounded by physics.



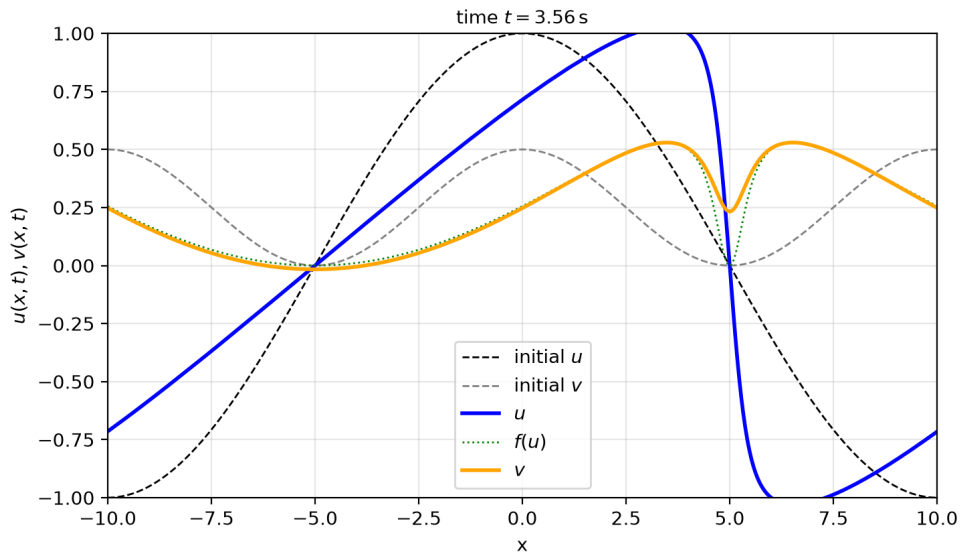
Stability Traps

- **Manifold Attraction:** Overfitting to physical diffusion $(-\varepsilon(aId - \mathbf{f}'(u)^2)\partial_{xx}u)$ instead of first-order fluxes.
- **Central Bias:** FCNs default to non-dissipative symmetric stencils, triggering high-frequency explosions under explicit time-stepping.

Constrained Optimization Framework

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{interp}} + \lambda_1 \mathcal{L}_{\text{additivity}} + \lambda_2 \mathcal{L}_{\text{stability}}$$

- $\mathcal{L}_{\text{interp}}$: Local error minimization against reference gradients.
- $\mathcal{L}_{\text{additivity}}$: Temporal semi-group coherence $(\Delta t/2 \times 2 = \Delta t)$.
- $\mathcal{L}_{\text{stability}}$: Spectral radius via Frozen Coefficient Fourier Analysis



Remark

The data-driven spatial operator is **not a closed-form exact numerical scheme**, but a structurally bounded optimization conditioned by the training distribution.

Core Research Directions

- **Function Space Adaptation:** Which underlying family of profiles is the network implicitly identifying during training? How does the local ReLU activation pattern map to this family?
- **Shock Specialization:** Can the network's inherent piecewise non-linearity act as a dynamic, data-driven flux limiter specifically optimized to capture sharp discontinuities without manual parameter tuning?



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